Exp. 9 Deployment of Attendance Management Application using Azure and GitHub

AIM

To deploy an Attendance management Application in Microsoft Azure using GitHub integration and enable continuous deployment.

TOOLS REQUIRED

- Microsoft Azure (Azure for Students Subscription)
- GitHub Repository
- Web Browser
- Node.js (Runtime Environment)

PROCEDURE

Step 1: Project Development

- Developed an **Attendance Management System** locally.
- Included features like:
  - User login
  - Generate QR
  - Mark attendance
- Tested the application locally before deployment.

Step 2: Upload to GitHub

- Created a GitHub repository:
  Project
- Uploaded project files using Git commands:

```
git init
git add .
git commit -m "Attendance App online version"
git remote add origin <GitHub URL>
git push -u origin main
```

- Verified repository upload successfully.

Step 3: Create Azure Web App

- Logged into Azure Portal.
- Created a **Web App** with the following configuration:
  - **App Name:** attendance-management-system
  - **Resource Group:** attendance-app
  - **Runtime:** Node.js
  - **Operating System:** Linux
  - **Region:** Central India
  - **App Service Plan:** F1 (Free Tier)

Step 4: Configure Deployment Center

- Opened the Web App in Azure.
- Navigated to **Deployment Center**.
- Selected:
  - Source: GitHub
  - Repository: Project
  - Branch: main
- Authorized GitHub access.

- Saved configuration.

Step 5: Continuous Deployment Setup

- Azure automatically created a **GitHub Actions workflow**.
- Deployment process:
    1. Code pushed to GitHub
    2. GitHub Actions triggered
    3. Azure builds and deploys app

Step 6: Deployment Verification

- Azure Web App status: **Running**
- Accessed the application using default URL:

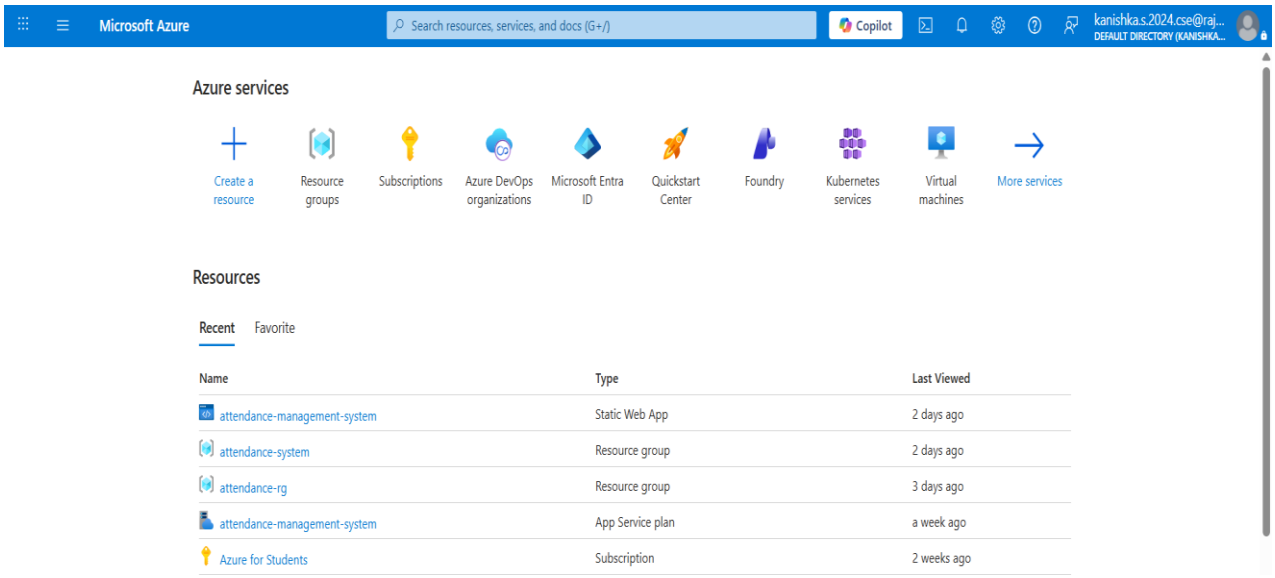
https://attendance-online-gugta9hkf2eadnh4.centralindia-01.azurewebsites.net

- Verified:
    - Attendance marked correctly
    - User interactions working
    - No runtime errors

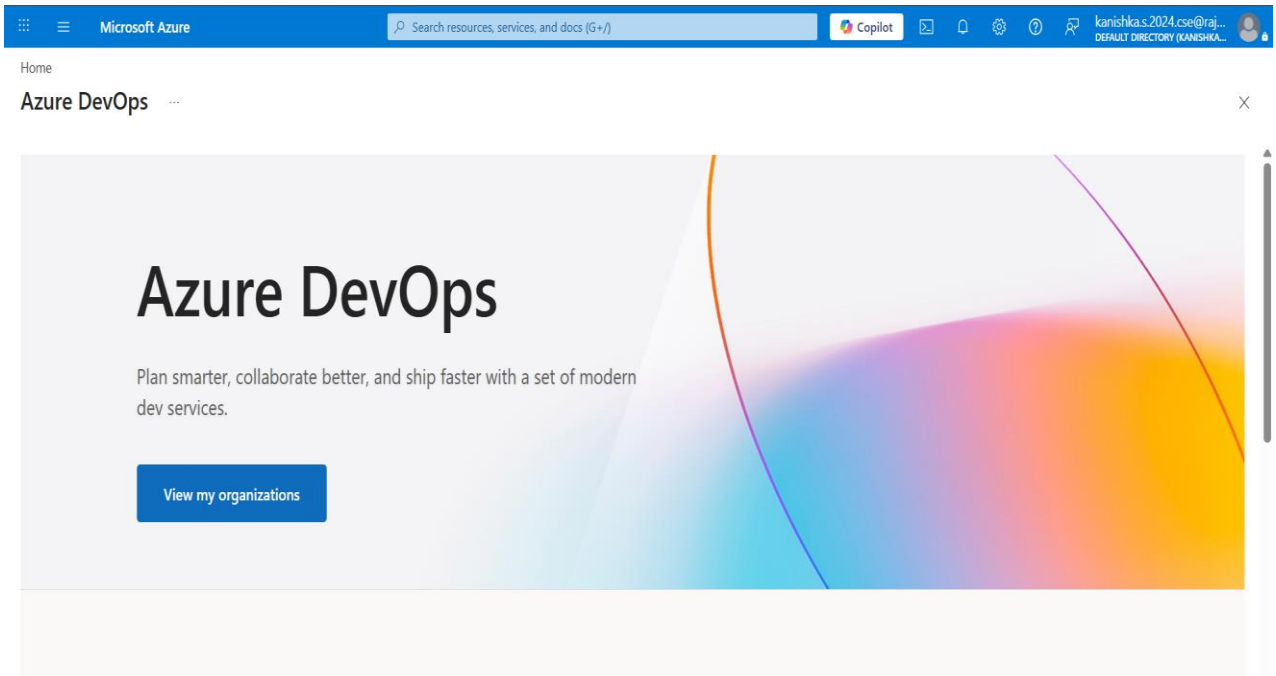
RESULT

The Attendance system Application was successfully deployed in Microsoft Azure. Continuous deployment was achieved using GitHub integration, enabling automatic updates whenever changes are pushed to the repository.

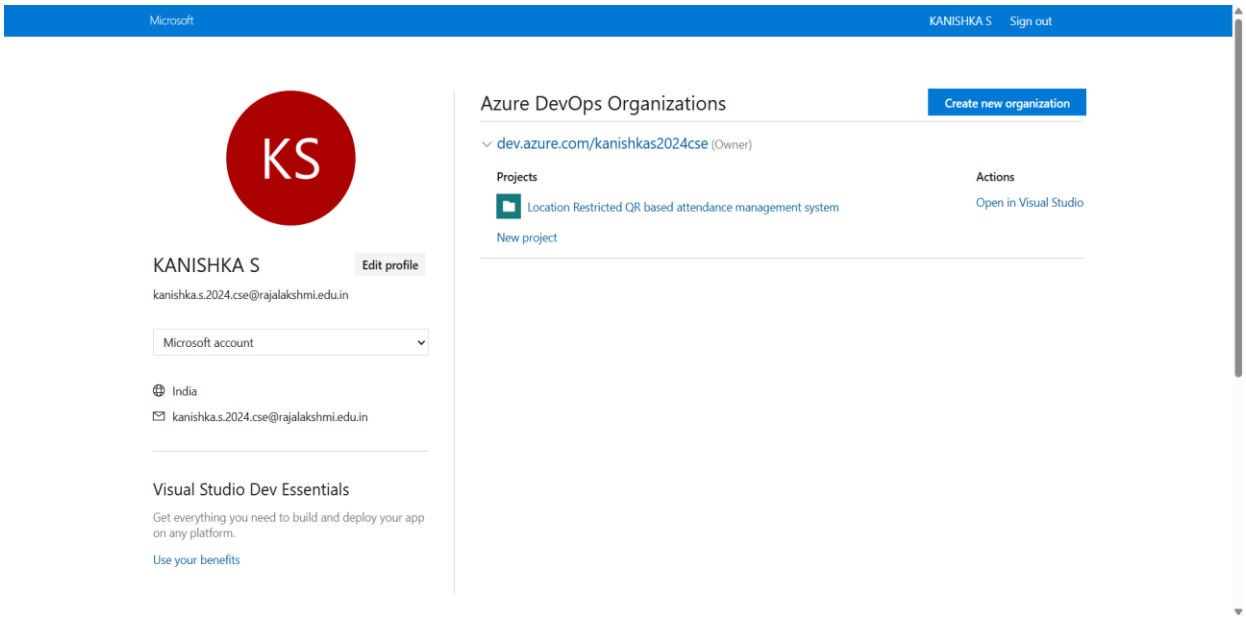
Microsoft Azure



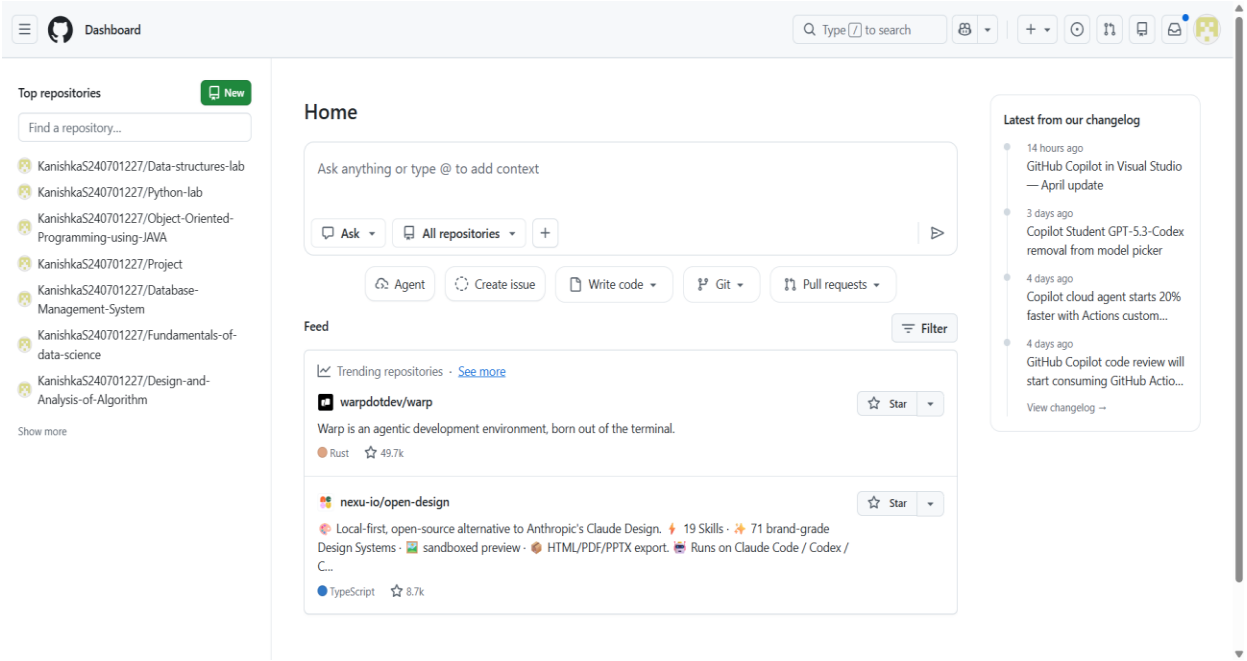
Azure DevOps Organizations



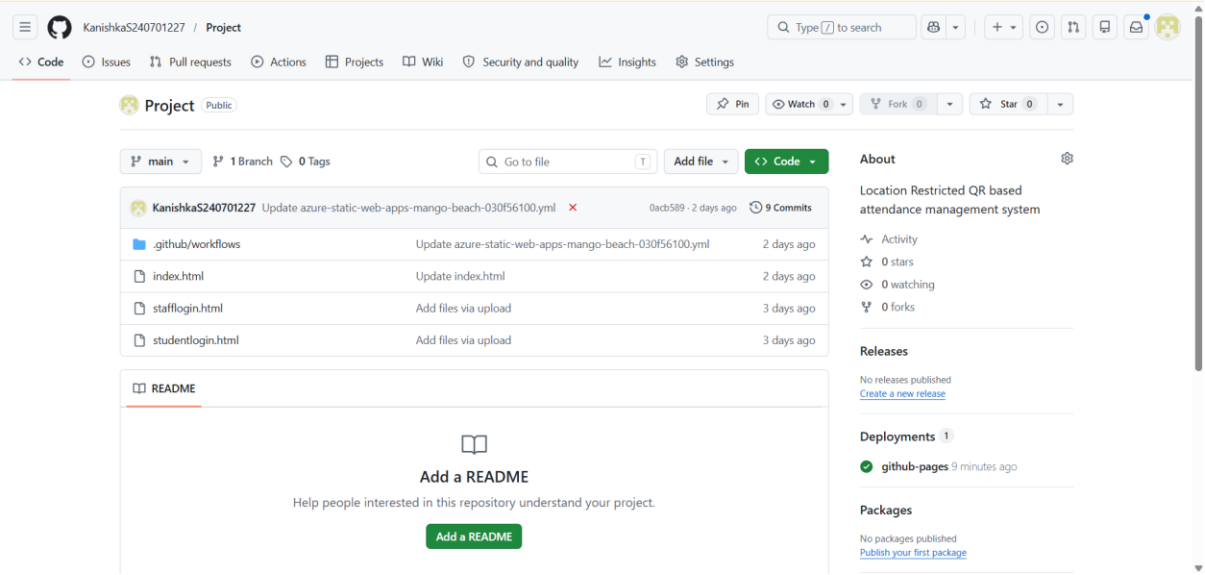
After clicking View my Organization



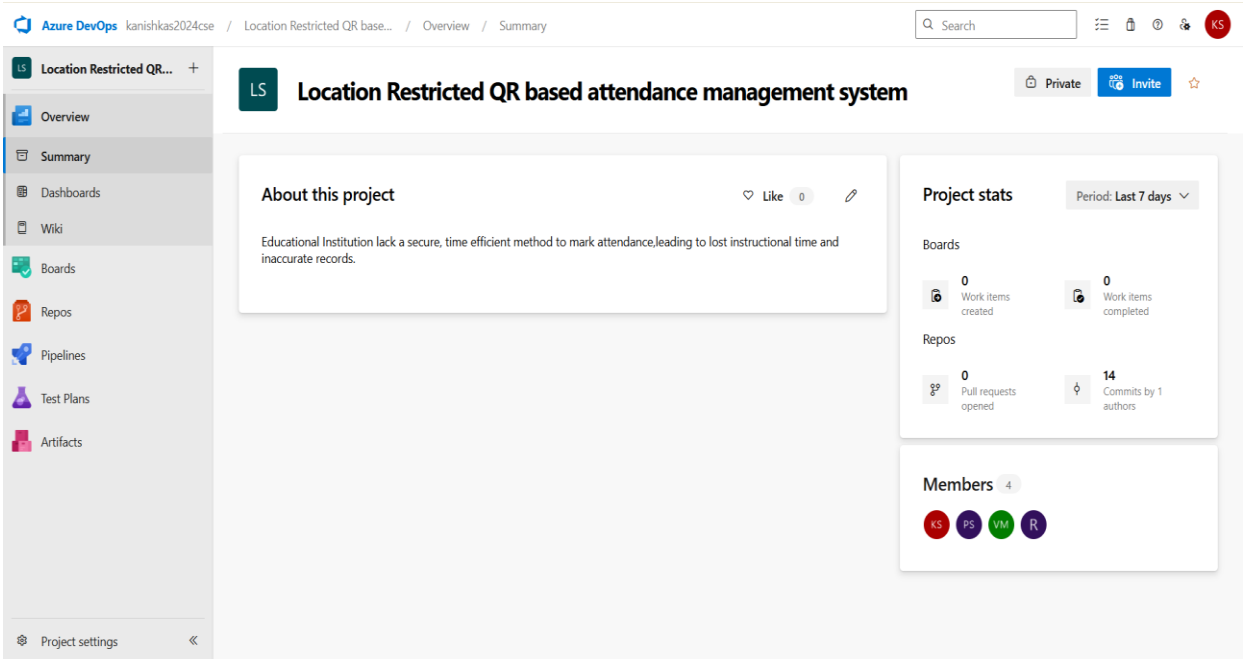
GitHub Repository



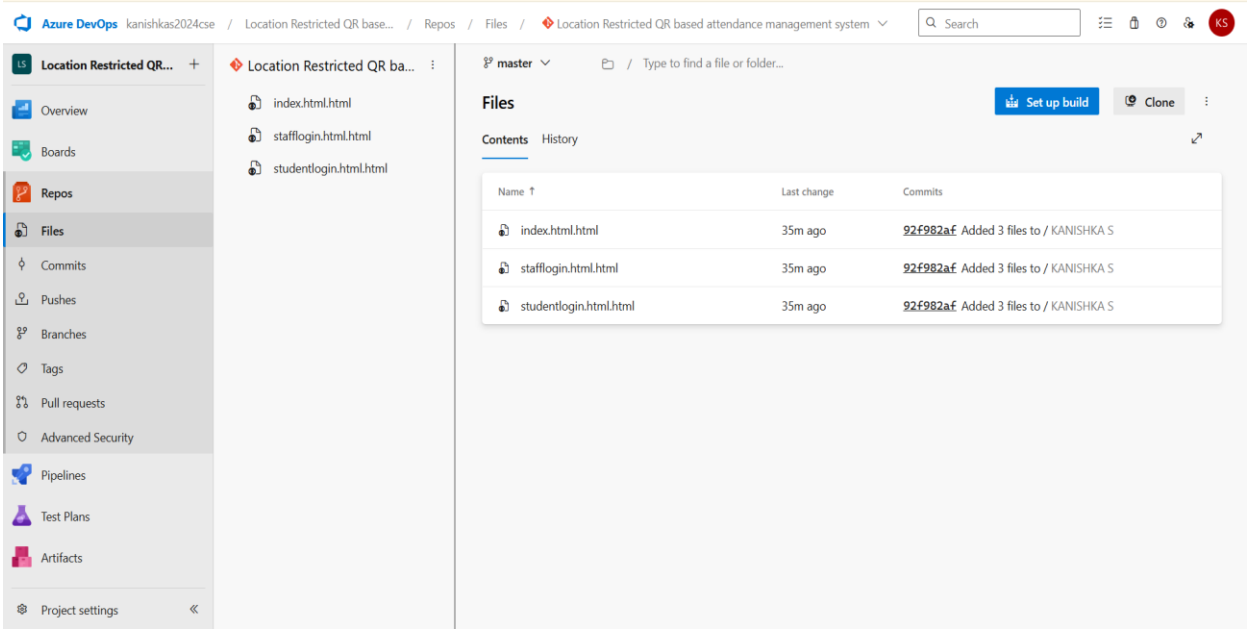
Attendance Management System Repo in GitHub

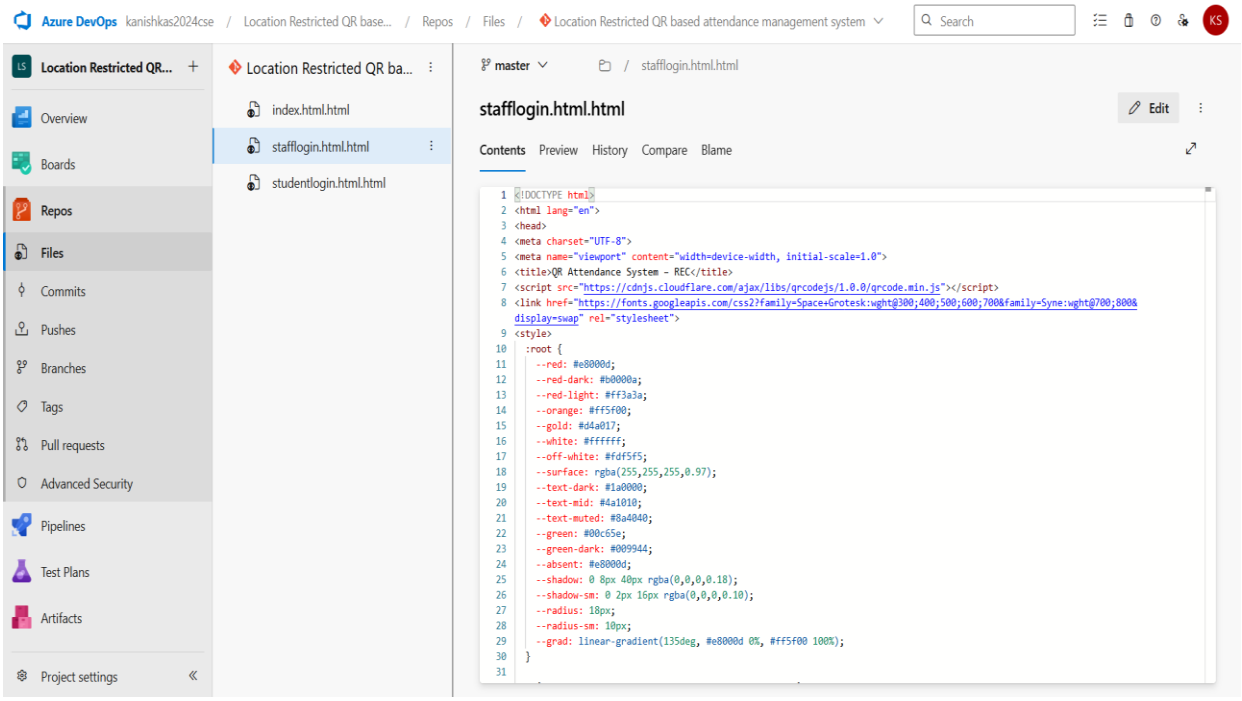


Attendance Management in Azure

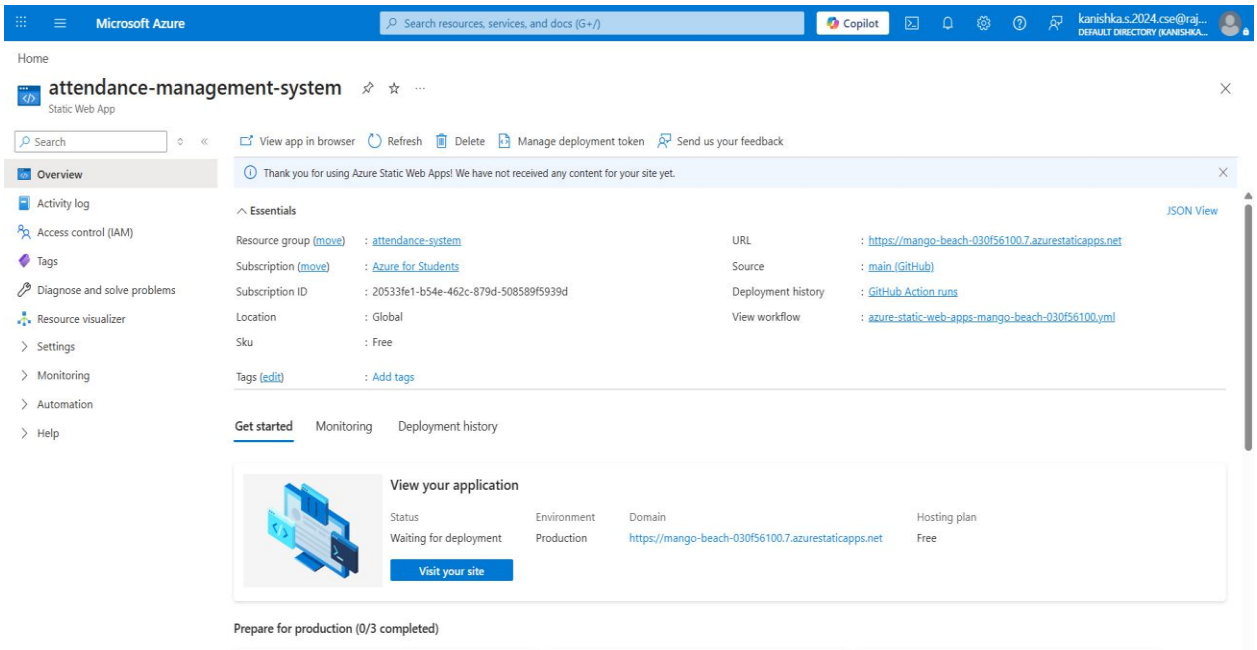


After Clicking Repo



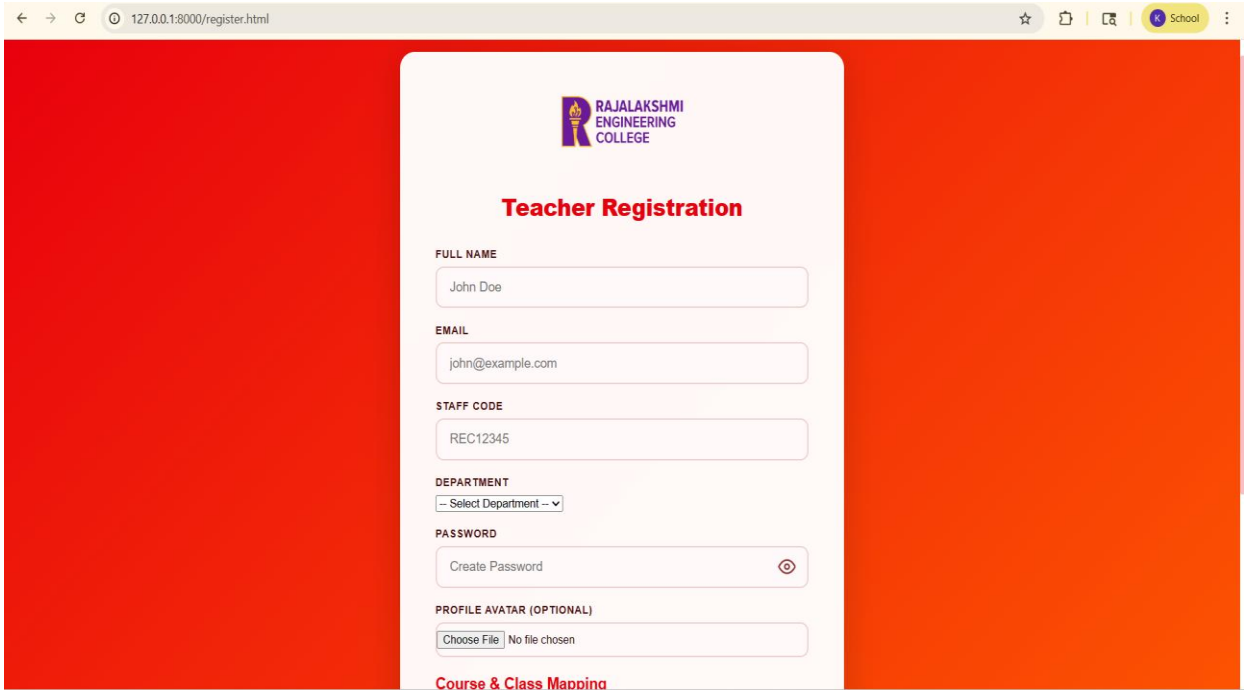


Web App

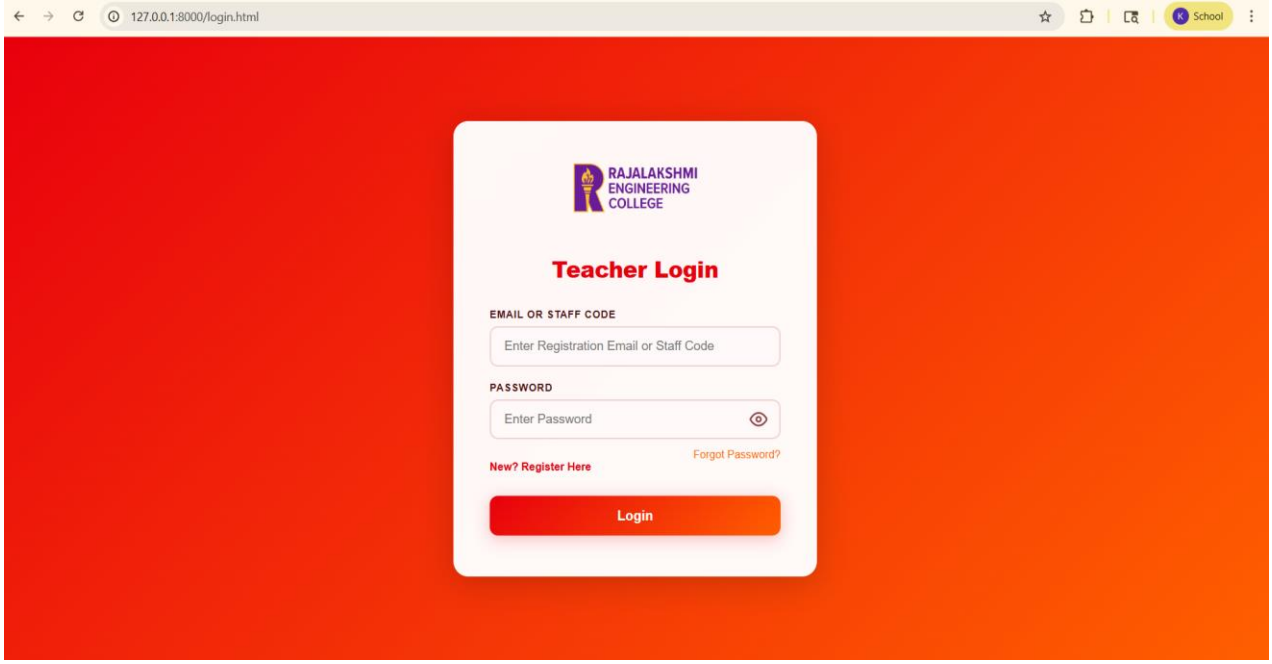


After clicking Browse : app runs in azure

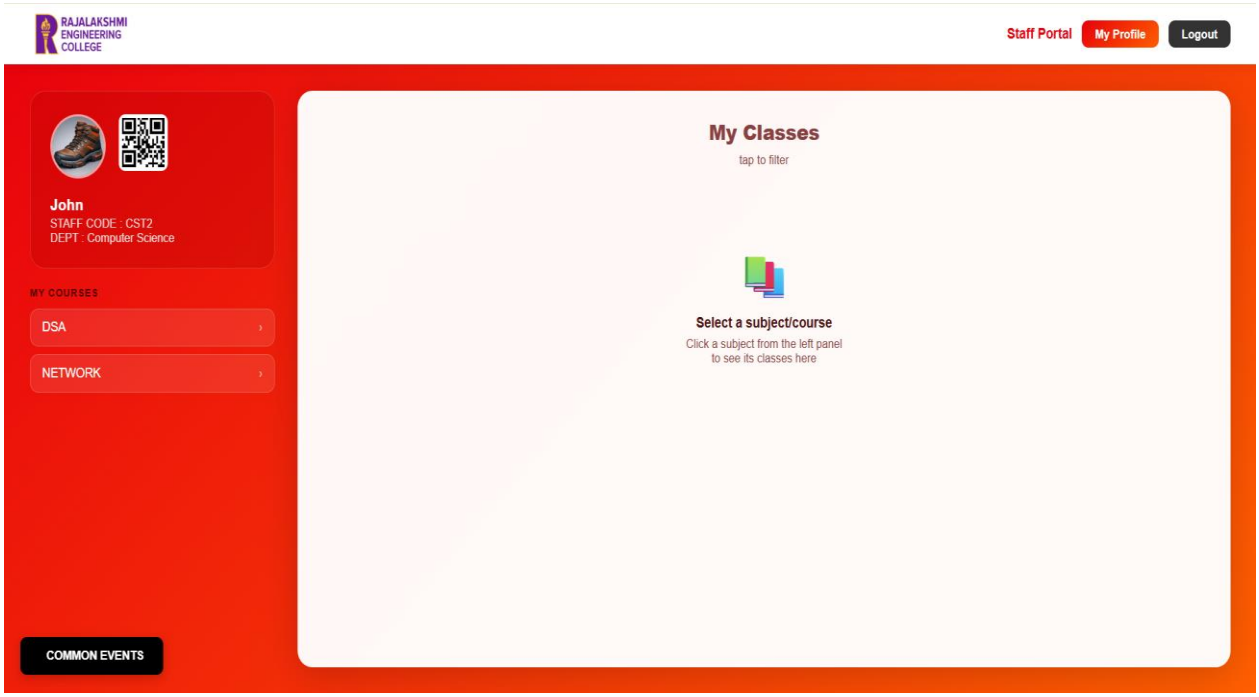
Register as Faculty



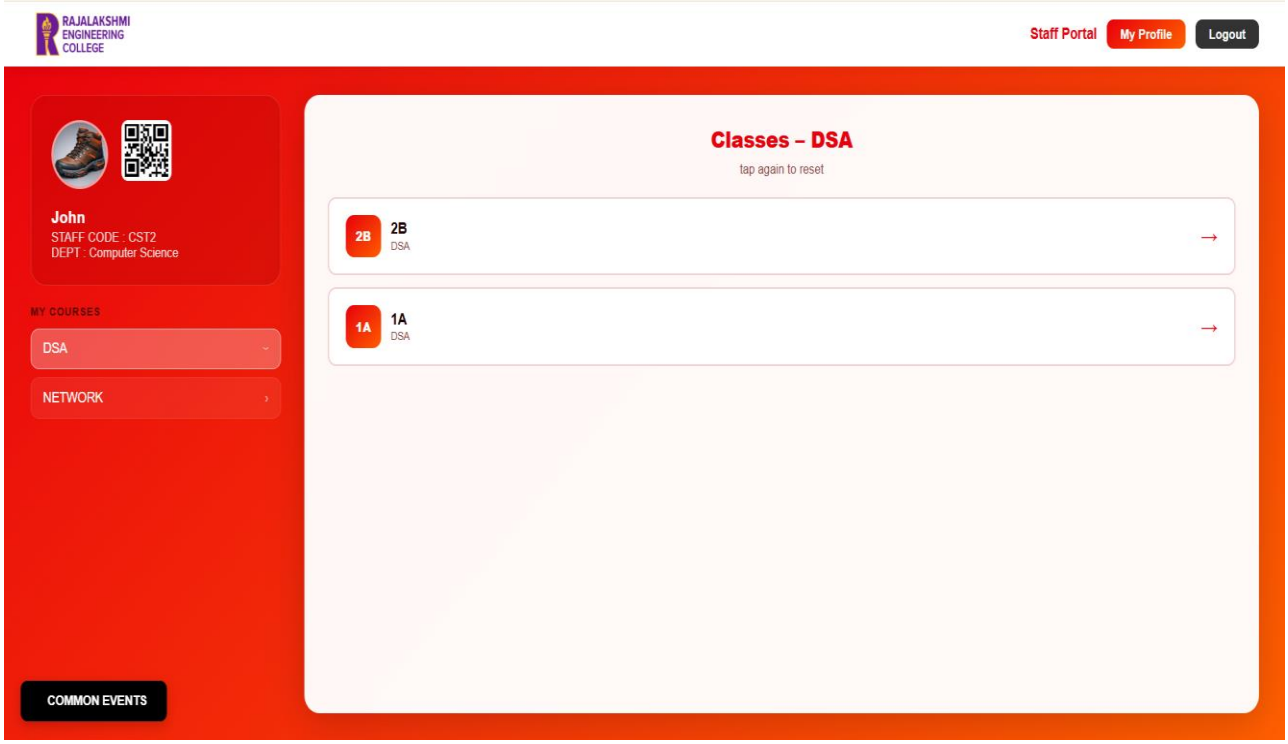
Login as Faculty



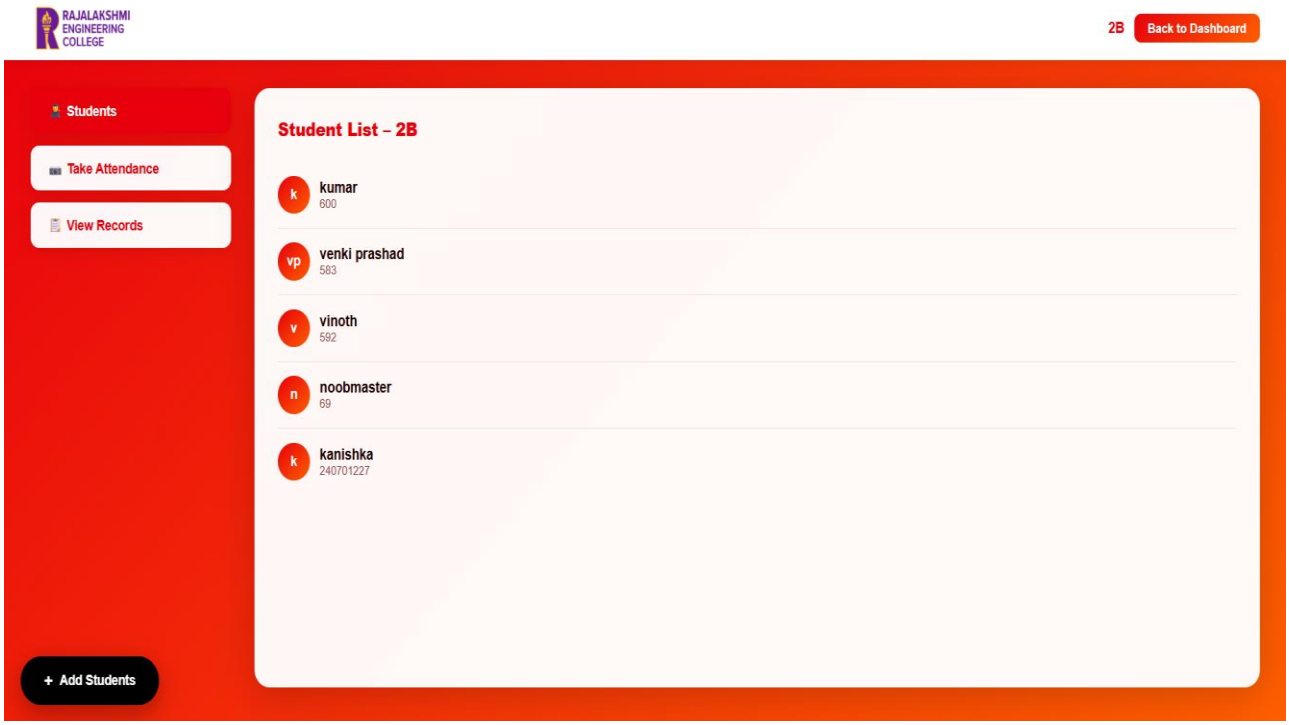
Faculty Dashboard



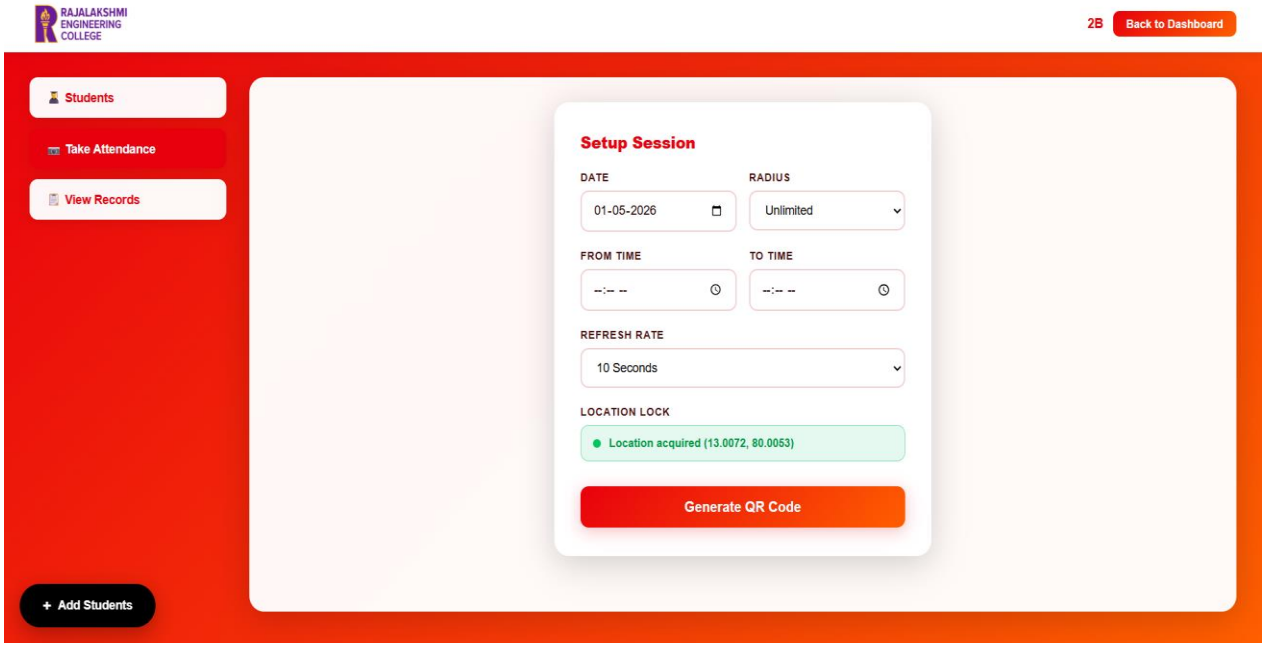
Select the Subject



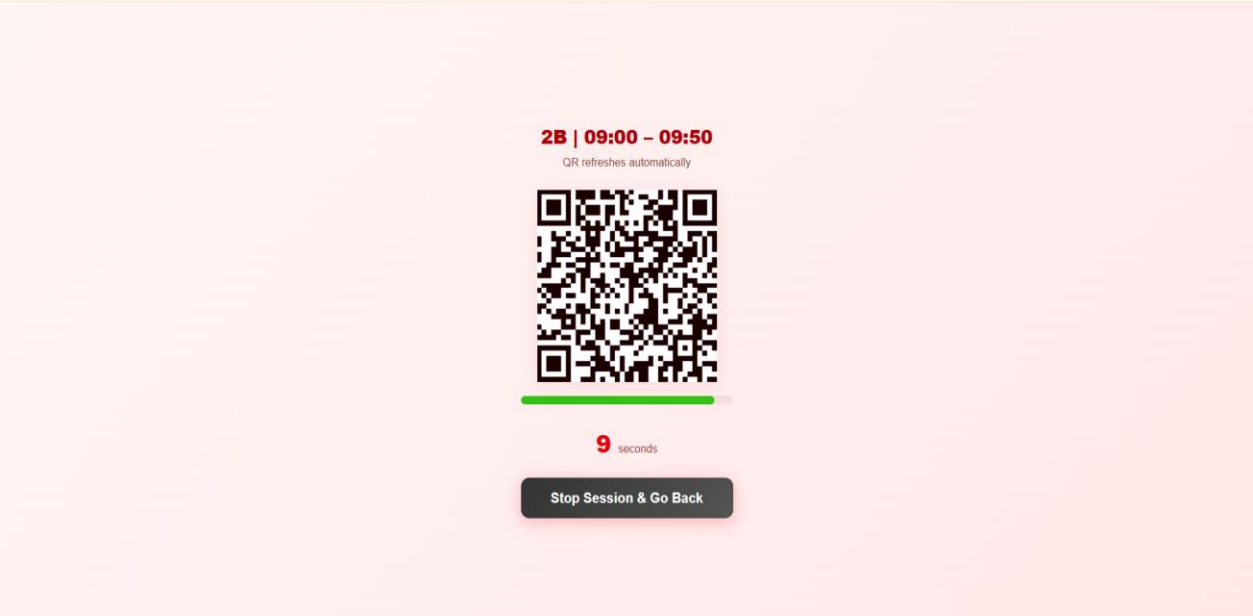
Select the Class



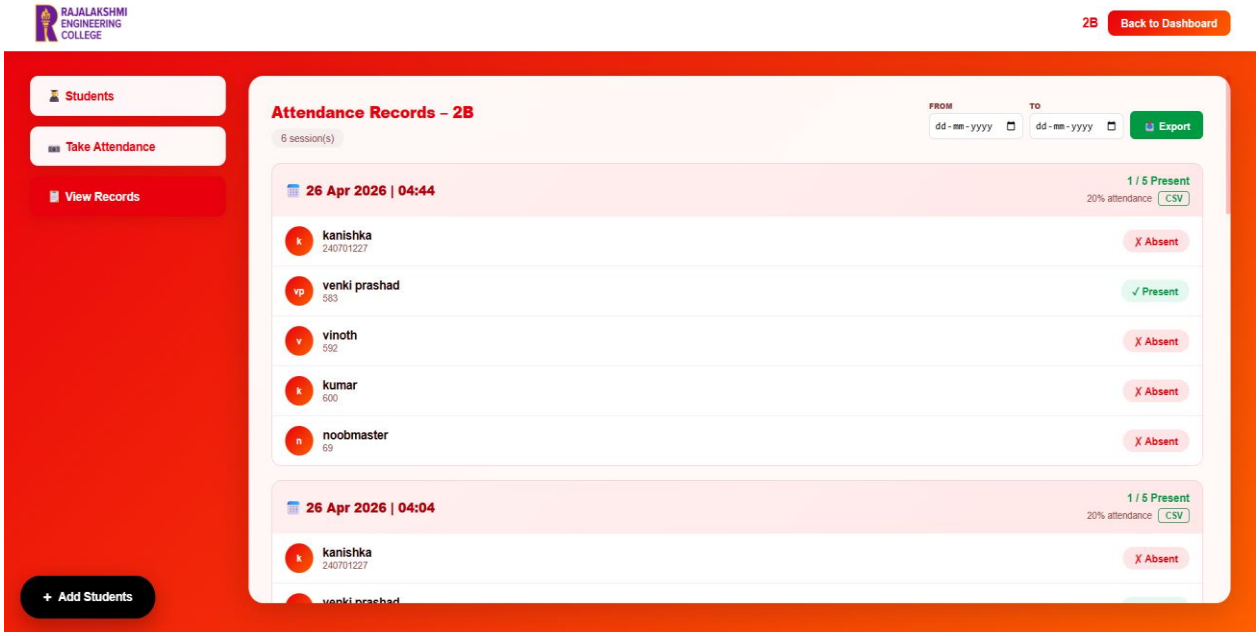
Take Attendance



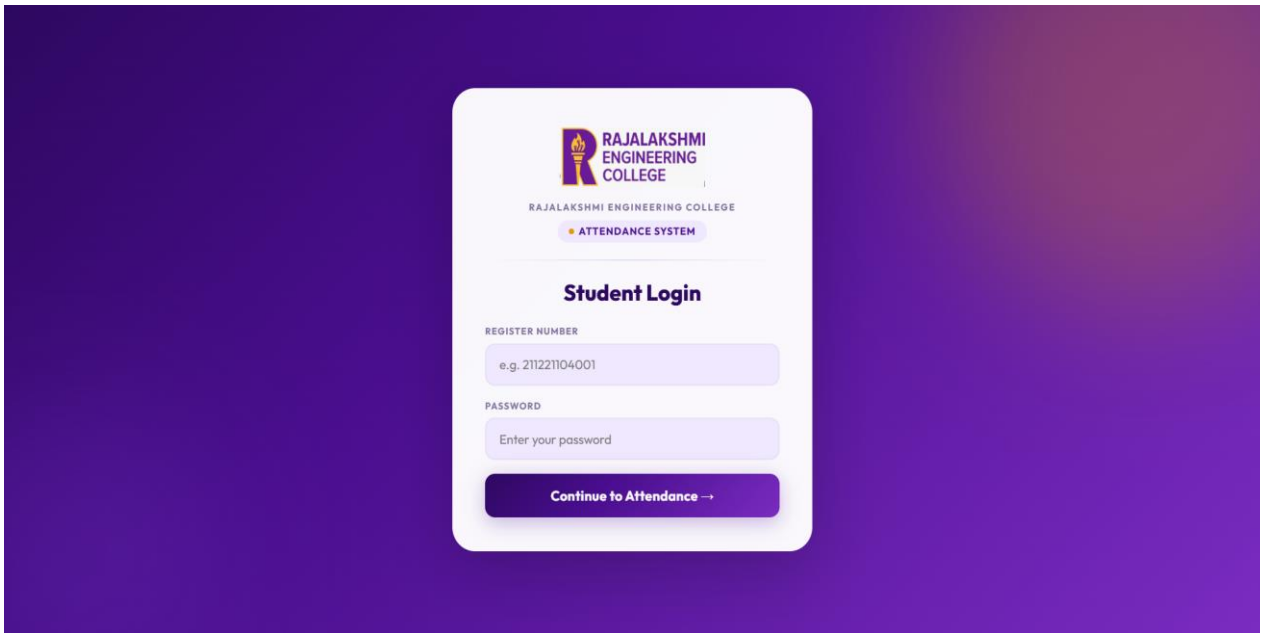
Generate QR



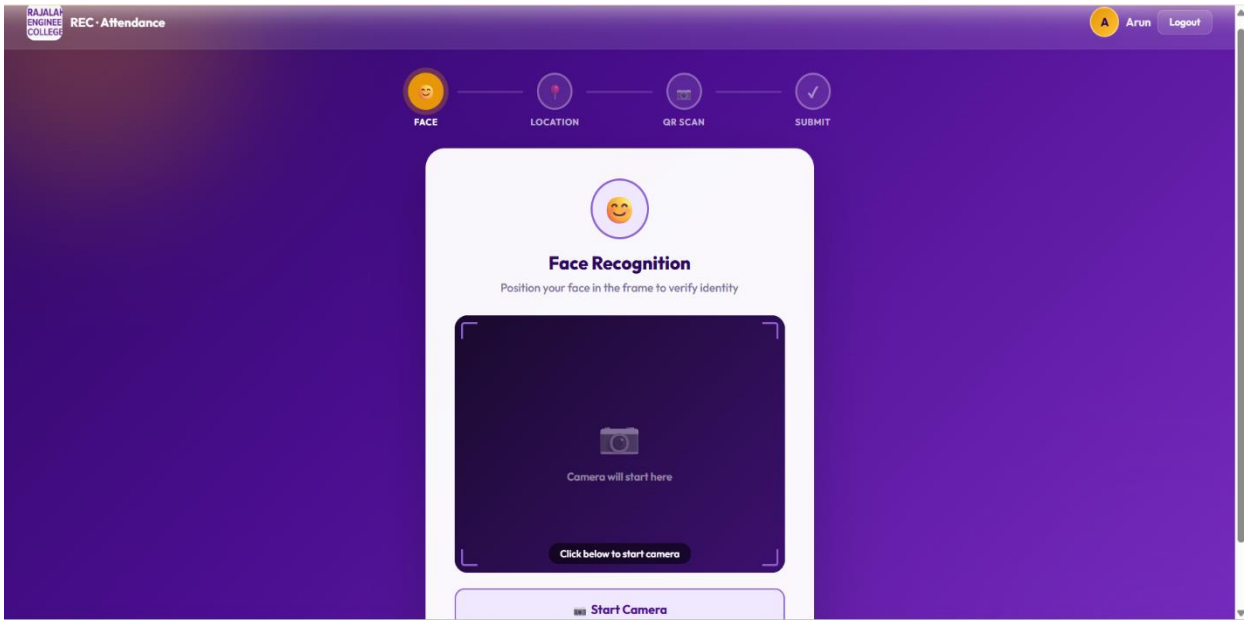
View Attendance



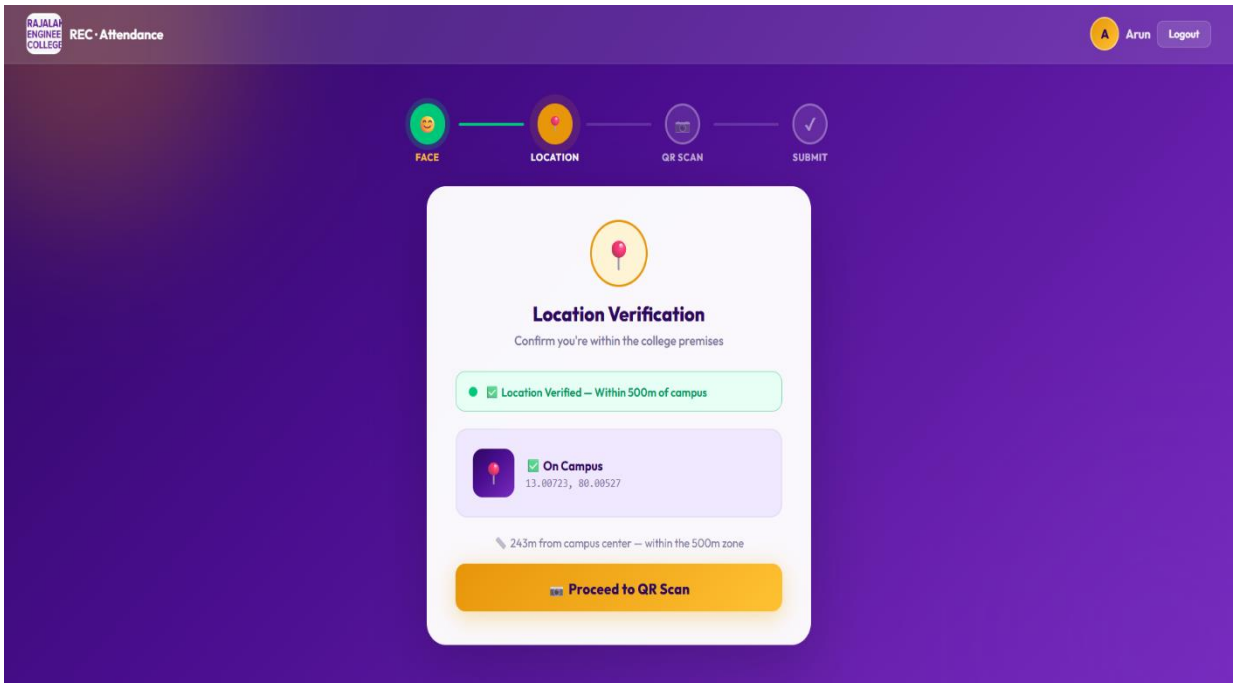
Student Login



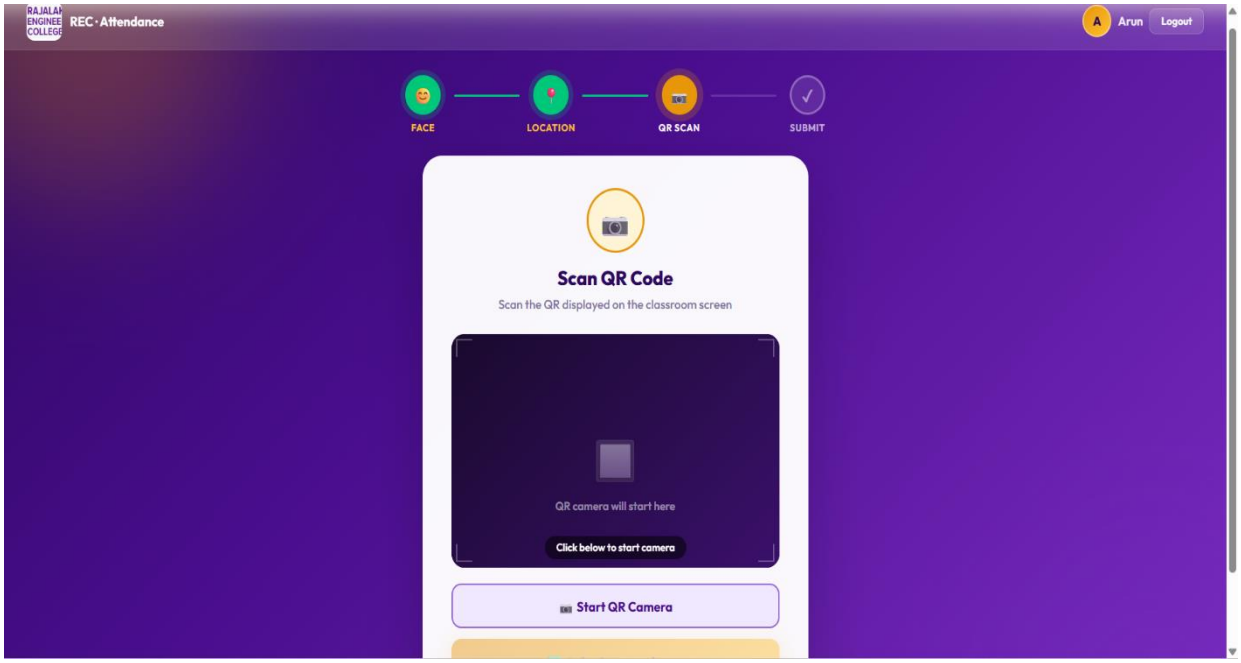
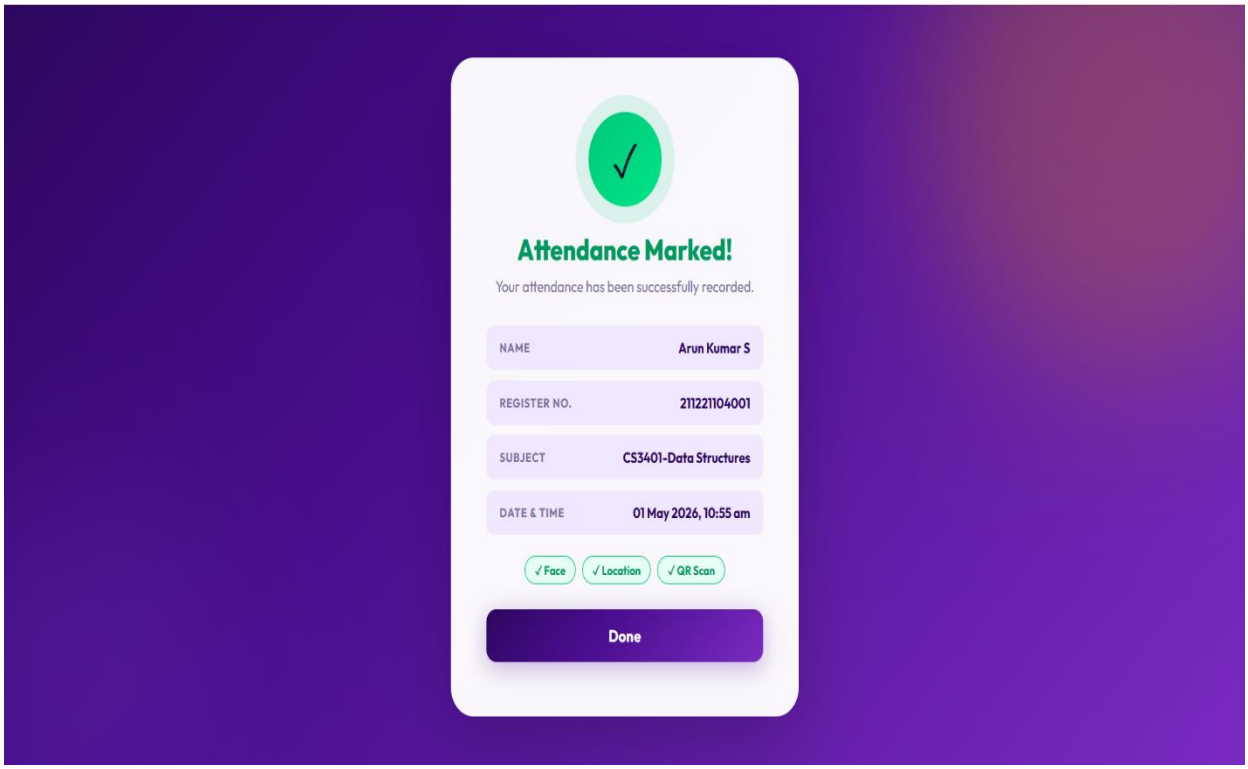
Face Recognition



Location Verification



Scan QR Code



Attendance Marked